

Introduction to Machine Learning

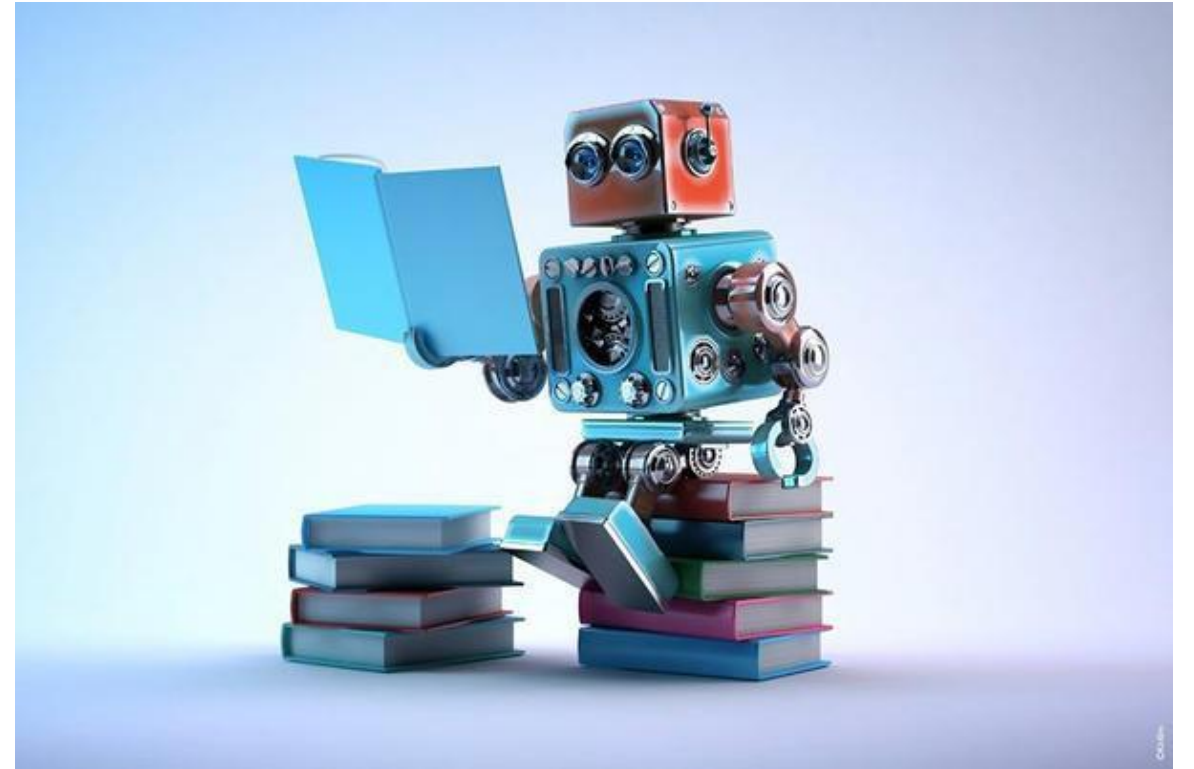
Serkan Ayvaz

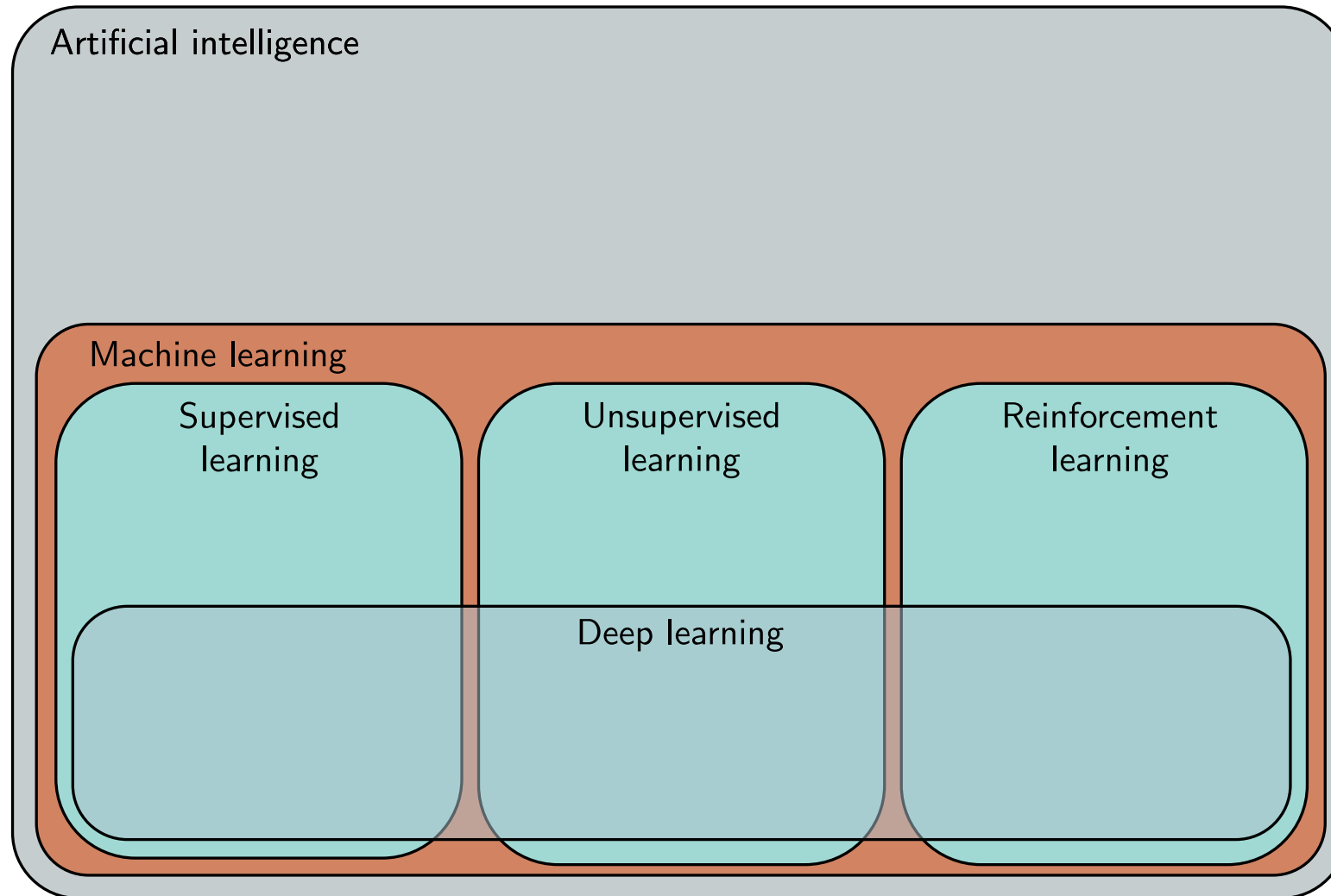
Machine learning

Definition

Getting a computer to do well on a task
without explicitly programming it

Improving performance on a task based on
experience

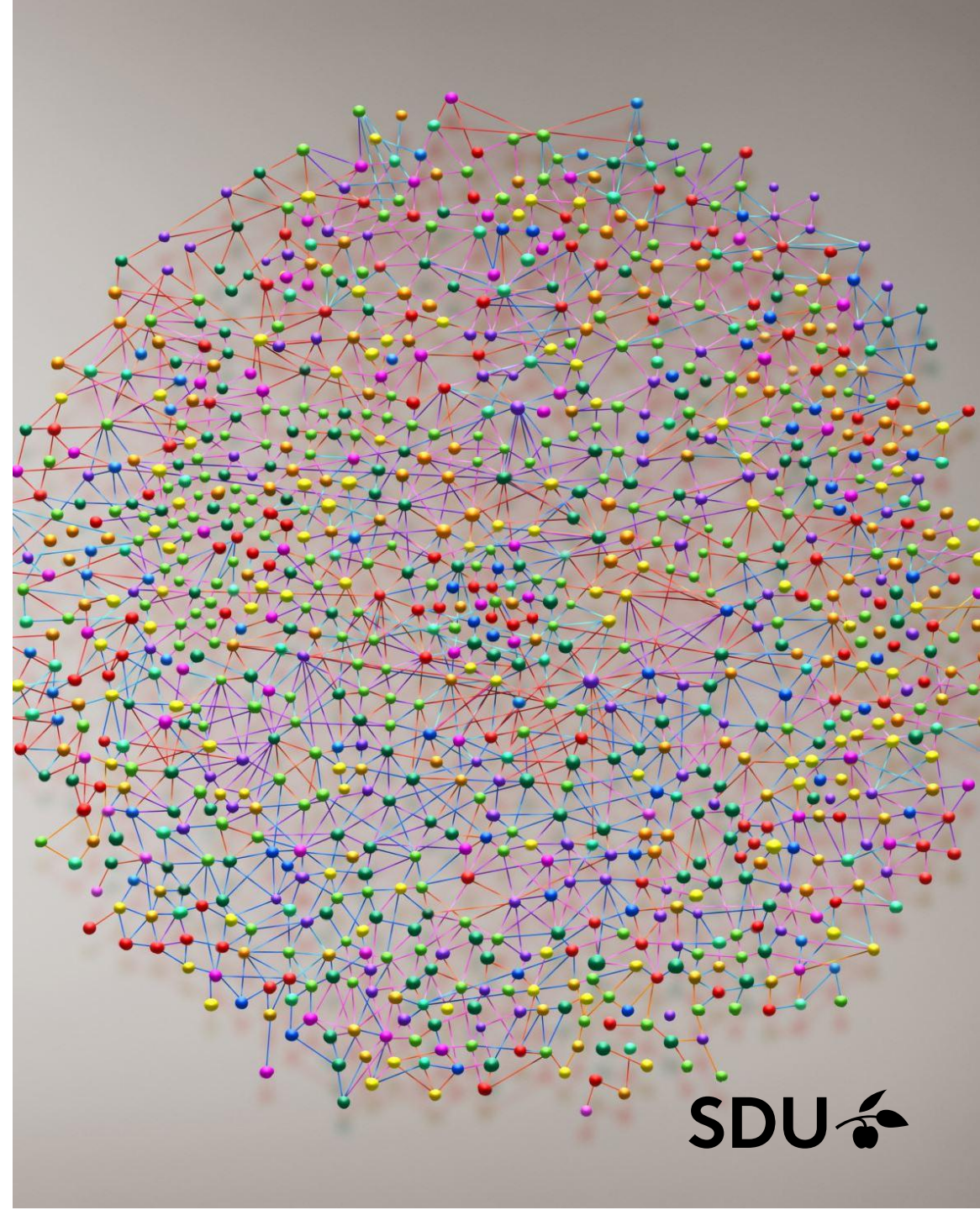




Unsupervised Learning

Learning about a dataset without labels

- Clustering
- Finding outliers
- Generating new examples
- Filling in missing data





Reinforcement learning

A set of **states**

A set of **actions**

A set of **rewards**

Goal: take actions to change the state so that you receive rewards

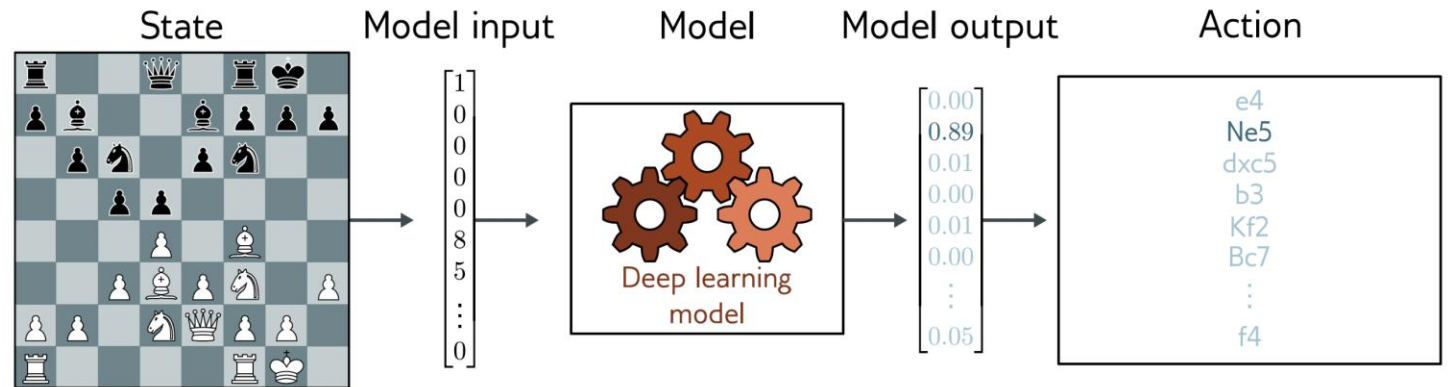
You don't receive any data – you have to explore the environment yourself to gather data as you go

Example: chess

States are valid states of
the chess board

Actions at a given time are
valid possible moves

Positive rewards for taking
pieces, negative rewards
for losing them





Why is RL difficult?

Stochastic

- Make the same move twice, the opponent might not do the same thing
- Rewards also stochastic (opponent does or doesn't take your piece)

Temporal credit assignment problem

- Did we get the reward because of this move? Or because we made good tactical decisions somewhere in the past?

Exploration-exploitation trade-off

- If we found a good opening, should we use this?
- Or should we try other things, hoping for something better?

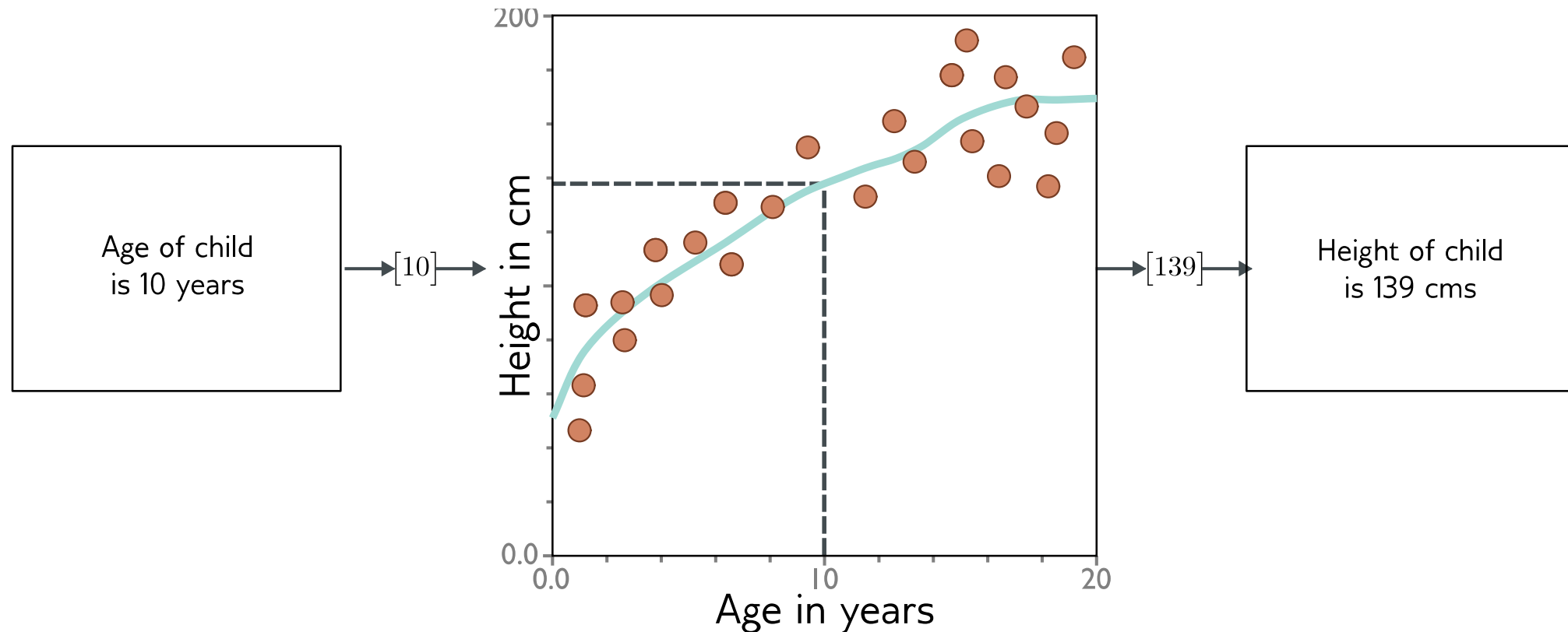


Supervised learning

Labeling: Define a **mapping** from input to output

Model learns the mapping from **paired input/output data** examples

What is a supervised learning model?



An equation relating input (age) to output (height)

Search through family of possible equations to find one that fits training data well



Terms

Regression = continuous numbers as output

Classification = discrete classes as output

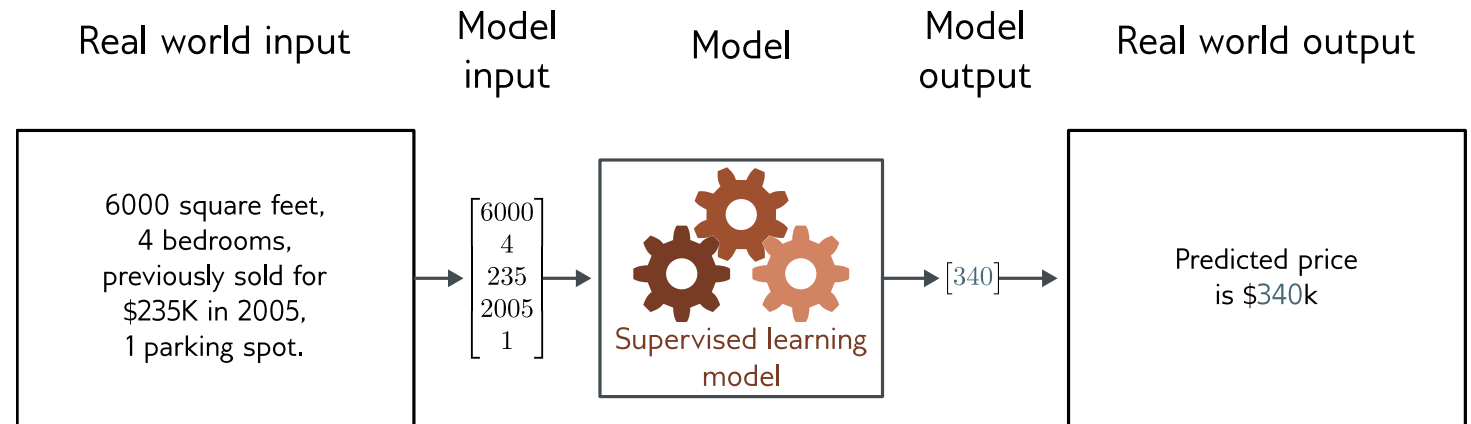
Two class and multiclass classification treated differently

Univariate = one output

Multivariate = more than one output

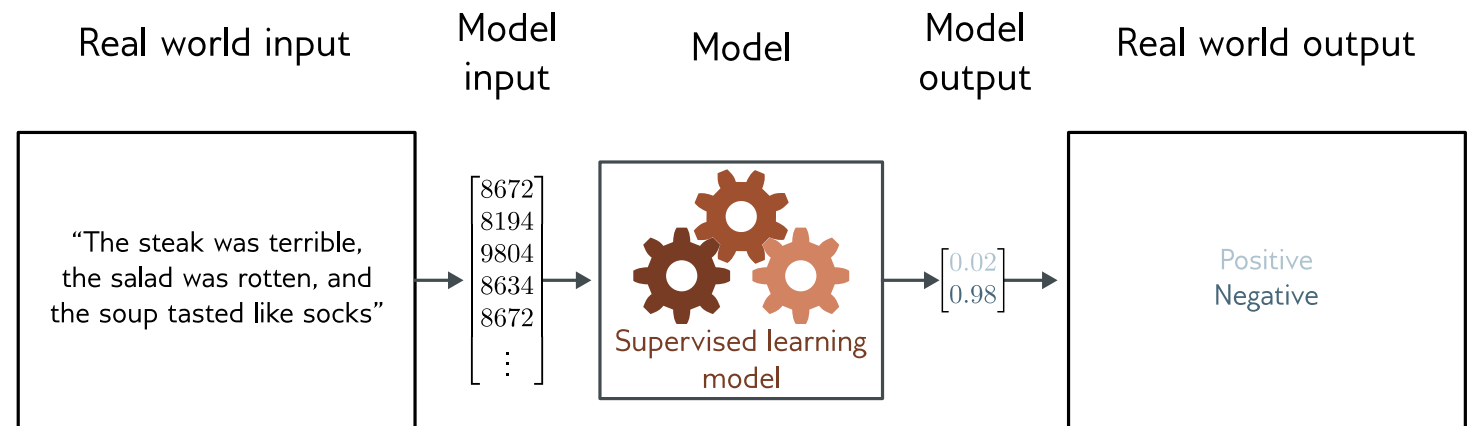
Regression

- Univariate regression problem (one output, real value)
- Fully connected network



Text classification

- Binary classification problem (two discrete classes)
- Transformer network



Music genre classification

- Multiclass classification problem (discrete classes, >2 possible values)
- Recurrent neural network (RNN)

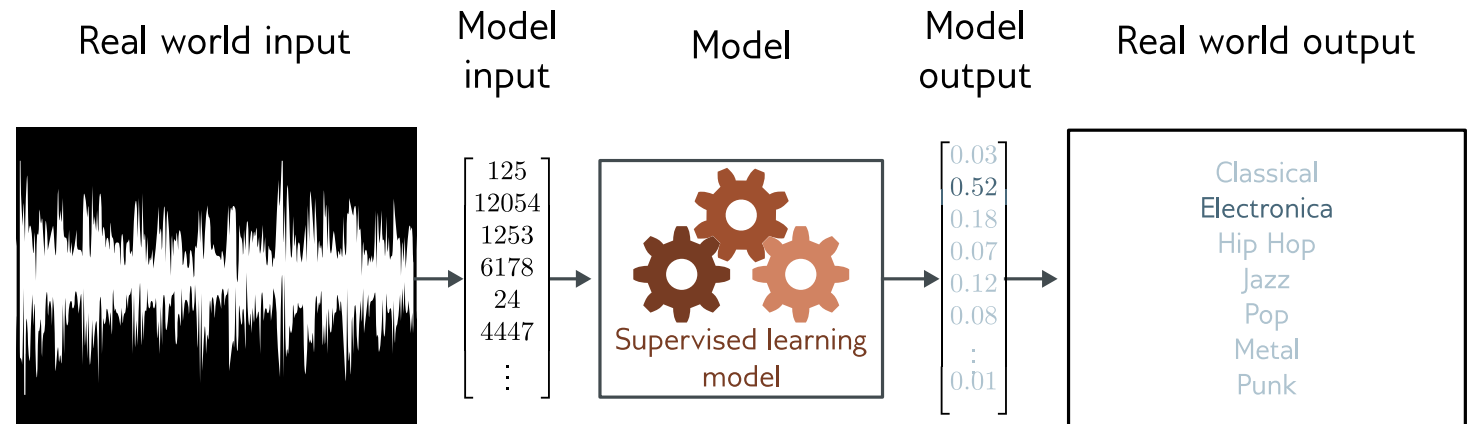


Image classification

- Multiclass classification problem (discrete classes, >2 possible classes)
- Convolutional network

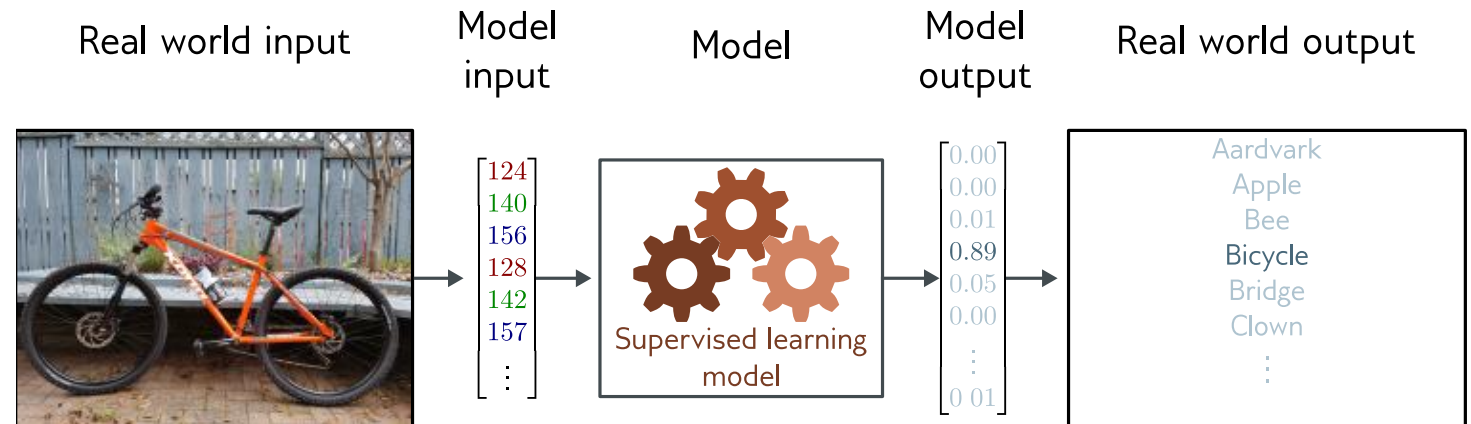
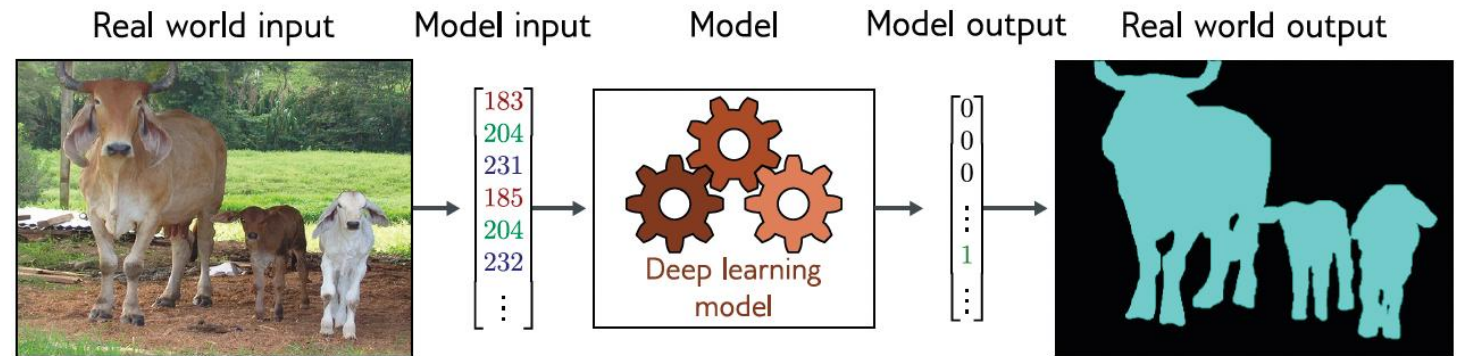
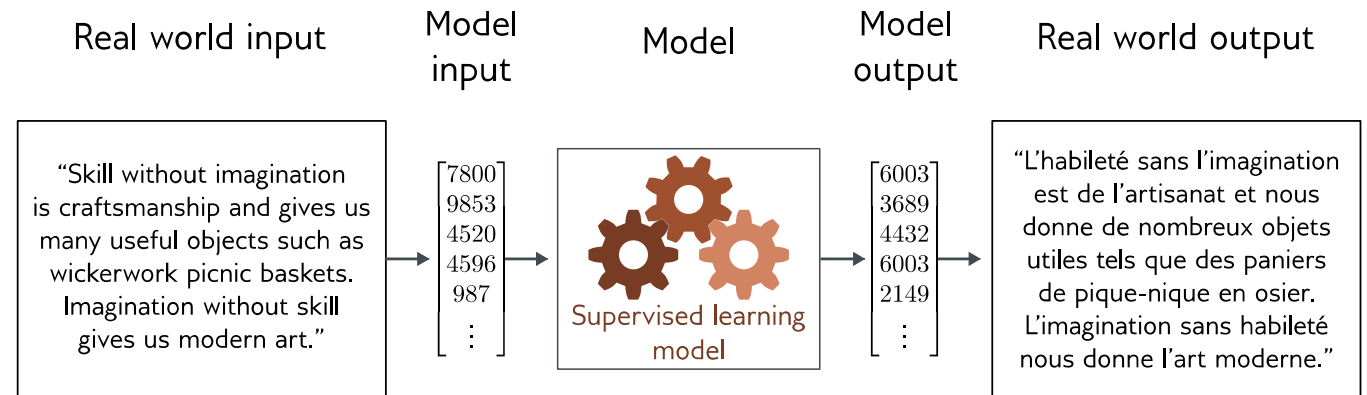


Image segmentation

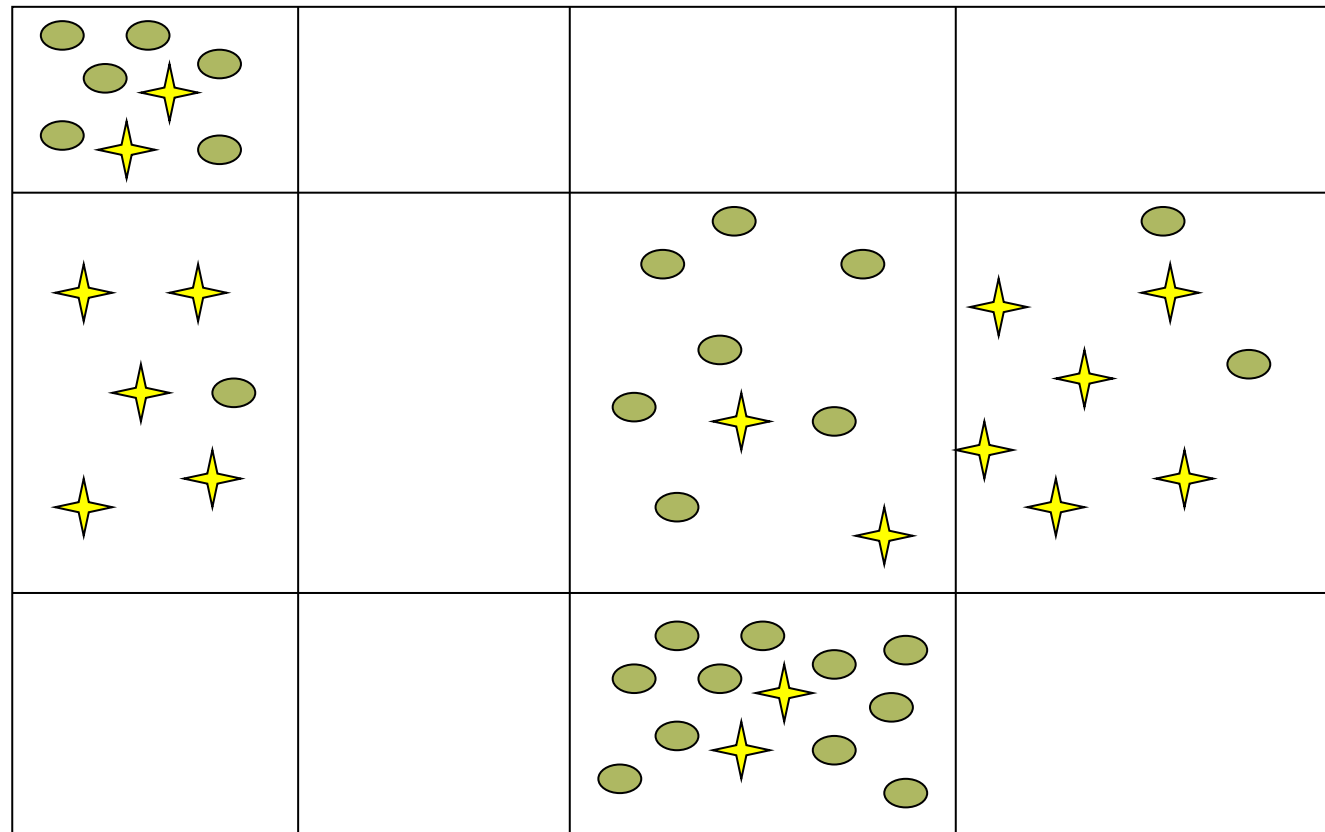
- Multivariate binary classification problem (many outputs, two discrete classes)
- Convolutional encoder-decoder network

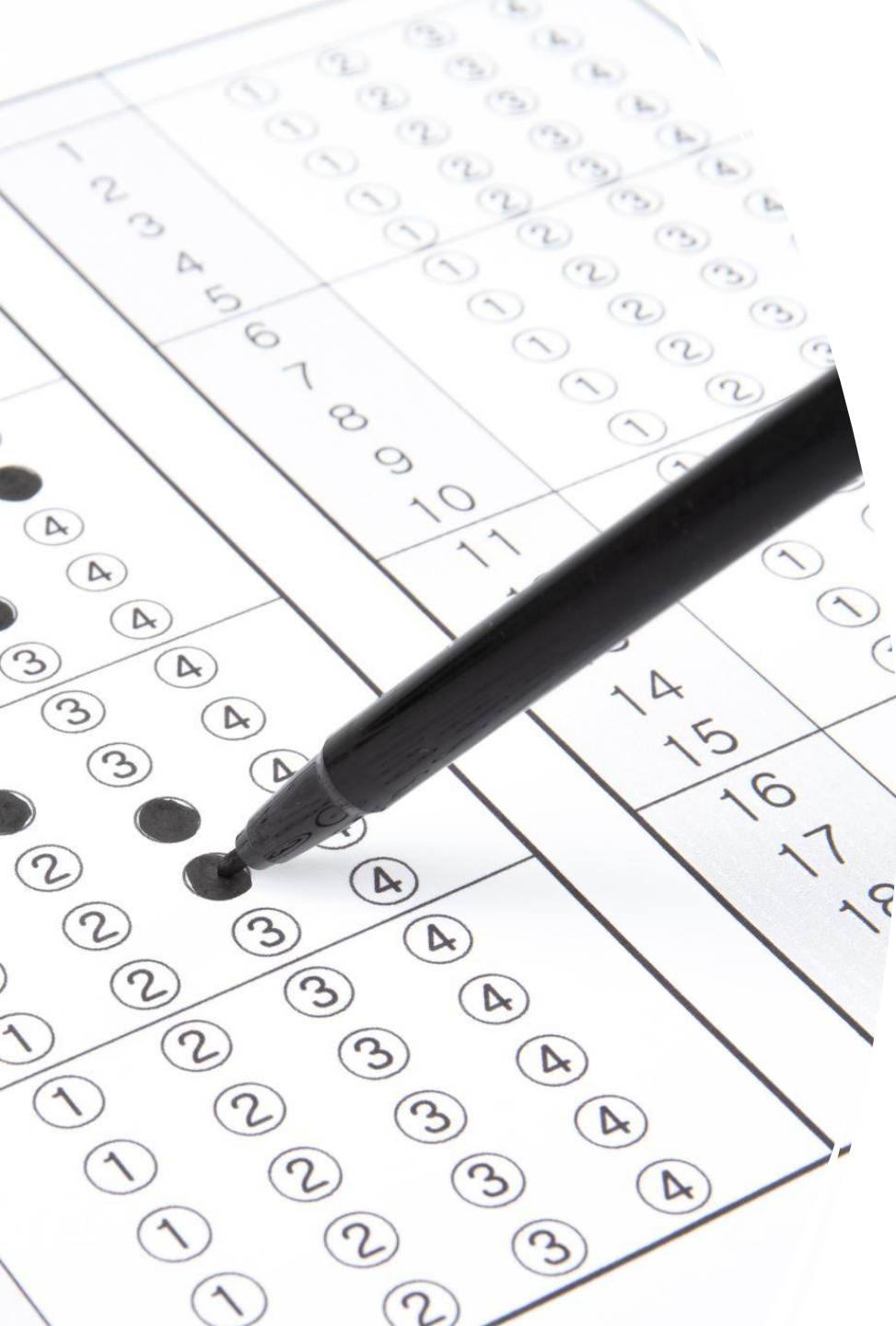


Translation



A Classification task





Classification: Definition

Given a collection of records (***training set***)

- Each record contains a set of ***attributes***, one of the attributes is the ***class***.

Find a ***model*** for class attribute as a function of the values of other attributes.

Goal: previously unseen records should be assigned a class as accurately as possible.

- A ***test set*** is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

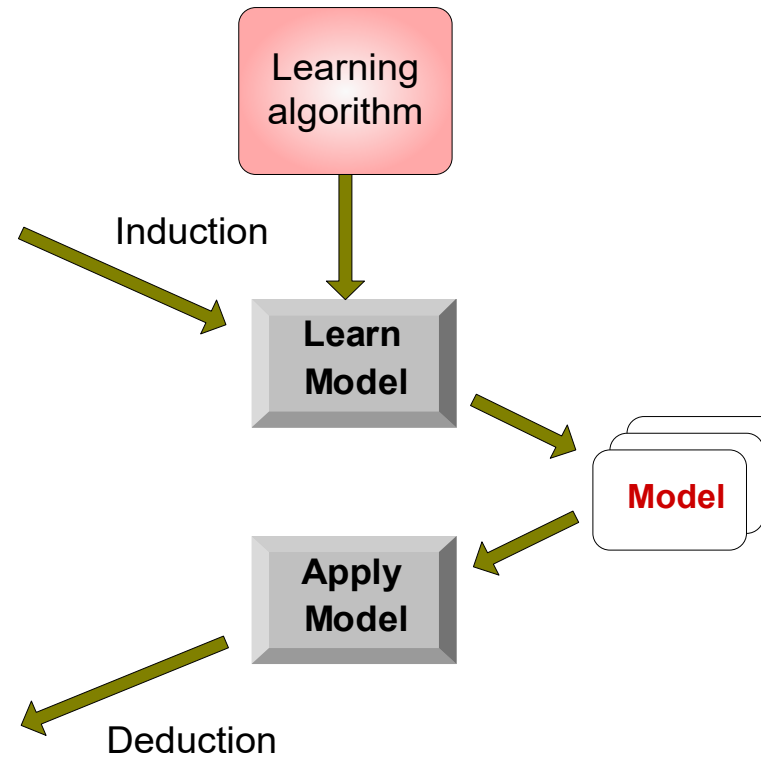
Illustrating Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Classification Techniques



Rule-based Methods



Distance-similarity based methods(Nearest-Neighbor)



Decision Tree based Methods



Probabilistic Methods(Naïve Bayes and Bayesian Belief Networks)



Support Vector Machines



Artificial Neural Networks



Ensemble Methods (Bagging - Boosting)

Random Forest
AdaBoost, CatBoost, XGBoost, etc.

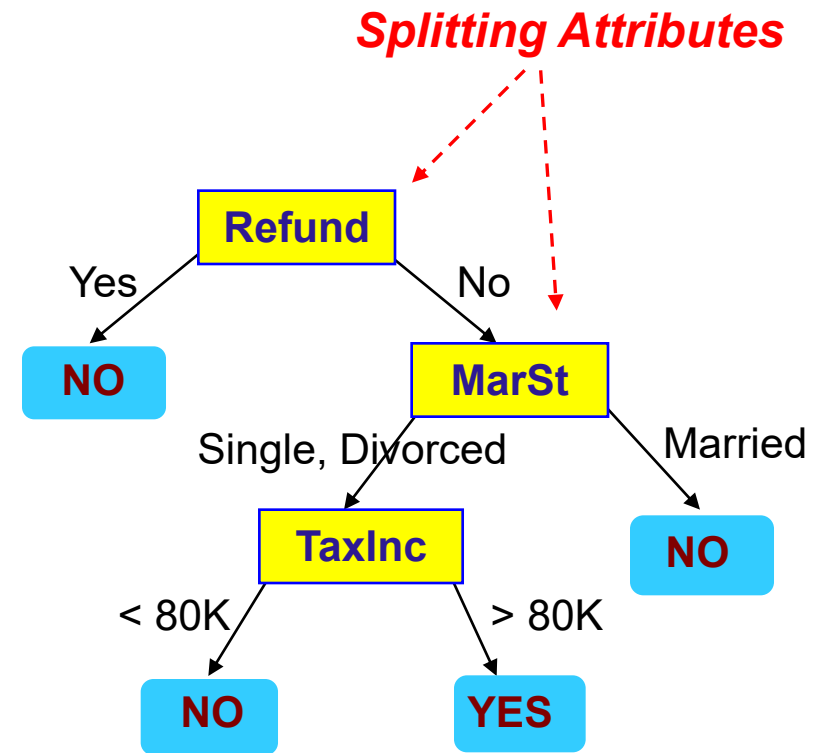
Decision Trees



Example of a Decision Tree

<i>Tid</i>	<i>Refund</i>	<i>Marital Status</i>	<i>Taxable Income</i>	<i>Cheat</i>
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Training Data

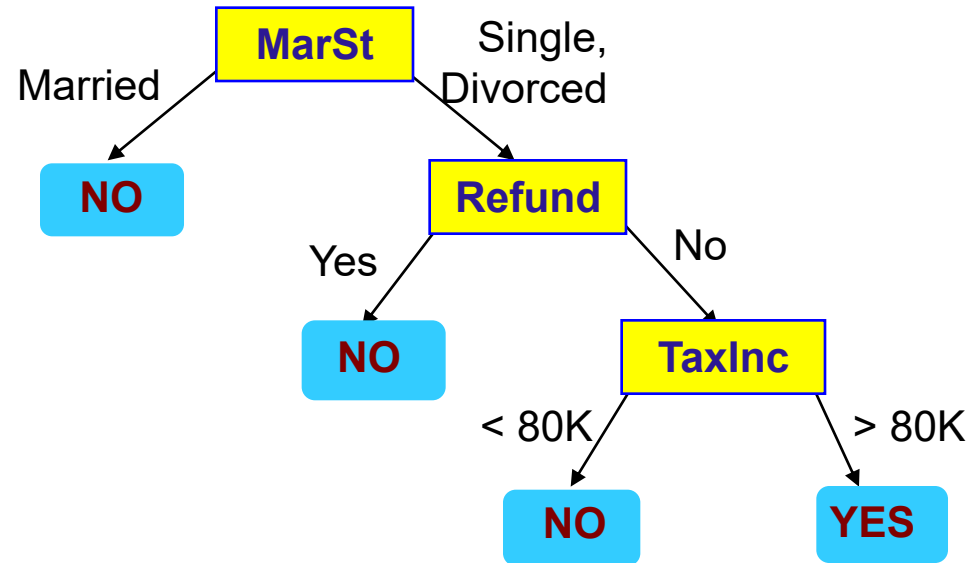


Model: Decision Tree

Another Example of Decision Tree

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

categorical
categorical
continuous
class



There could be more than one tree that fits the same data!

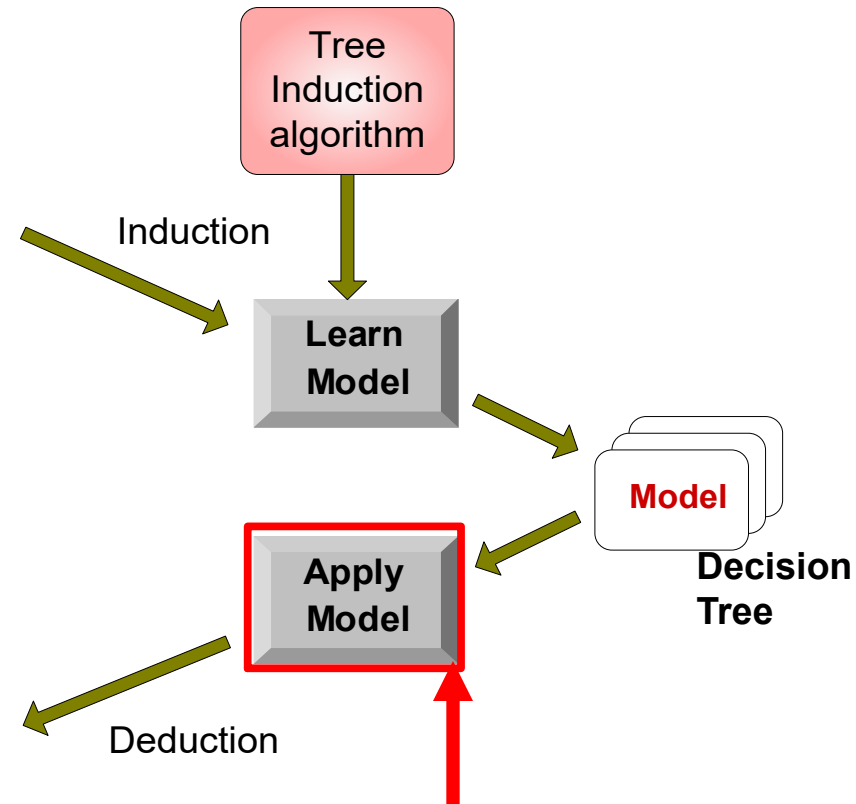
Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

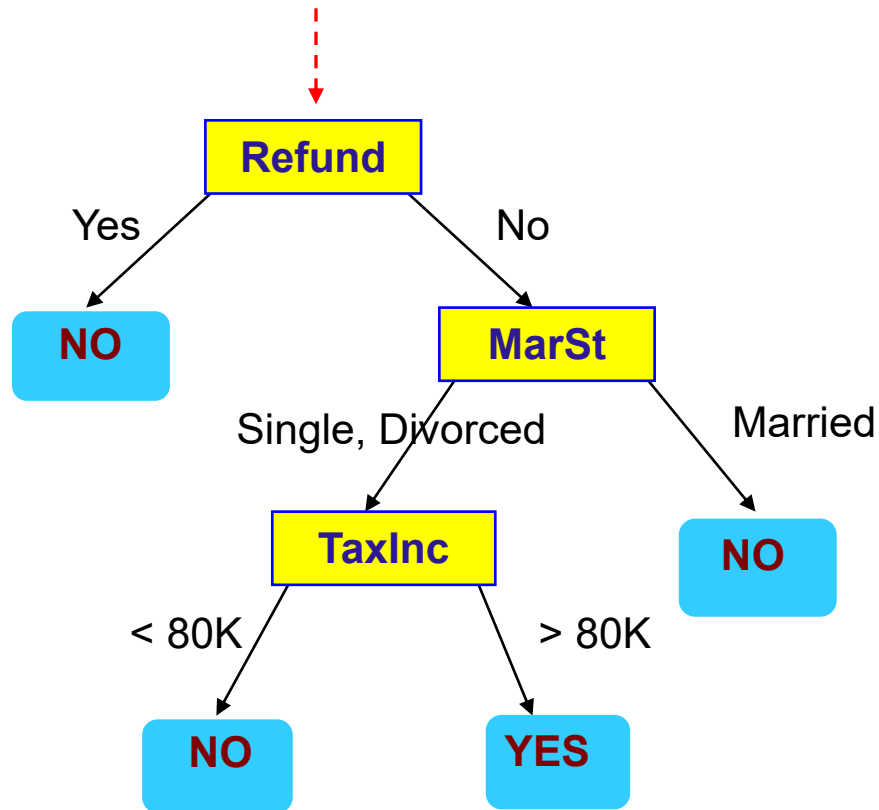
Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Apply Model to Test Data

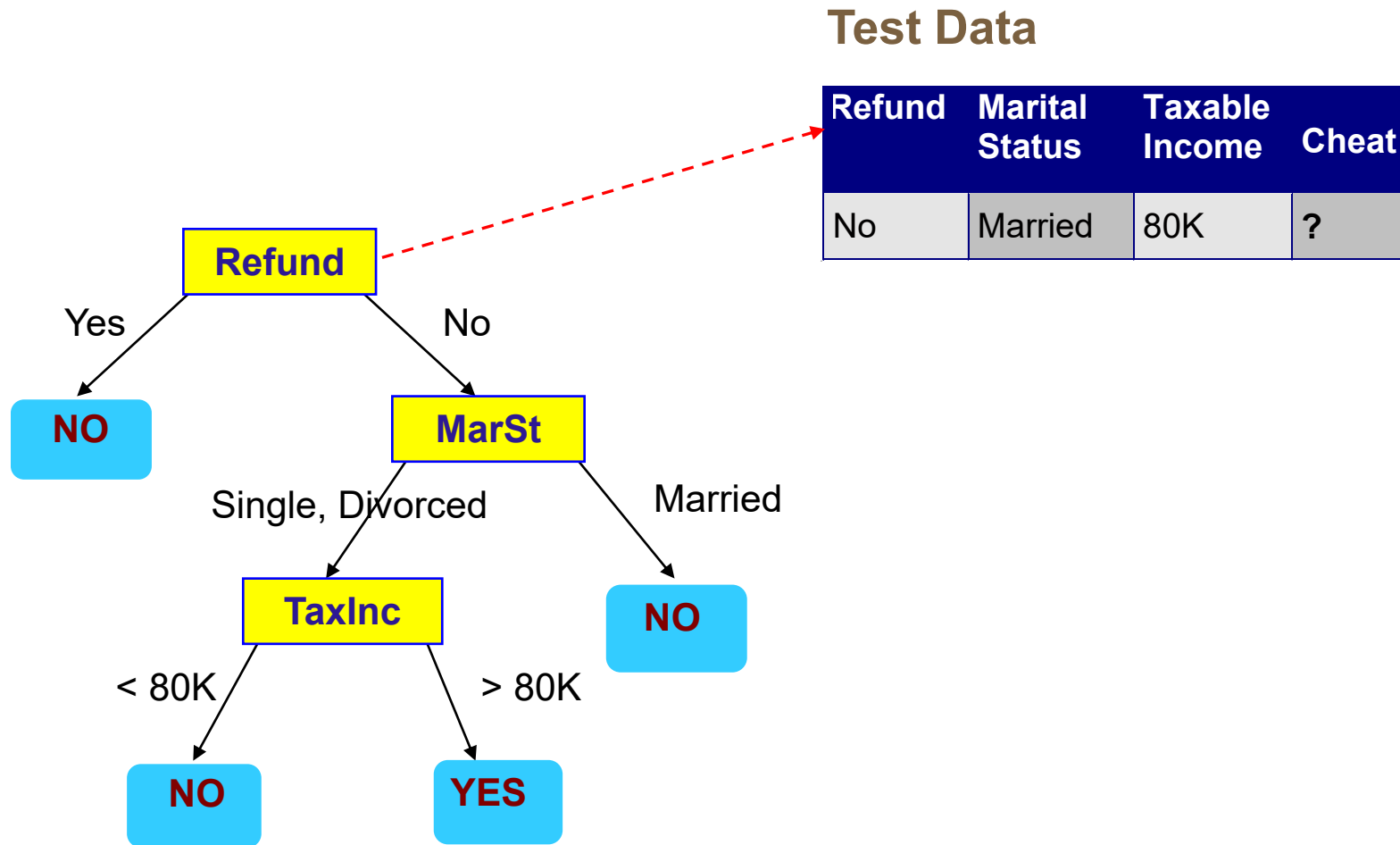
Start from the root of tree.



Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?

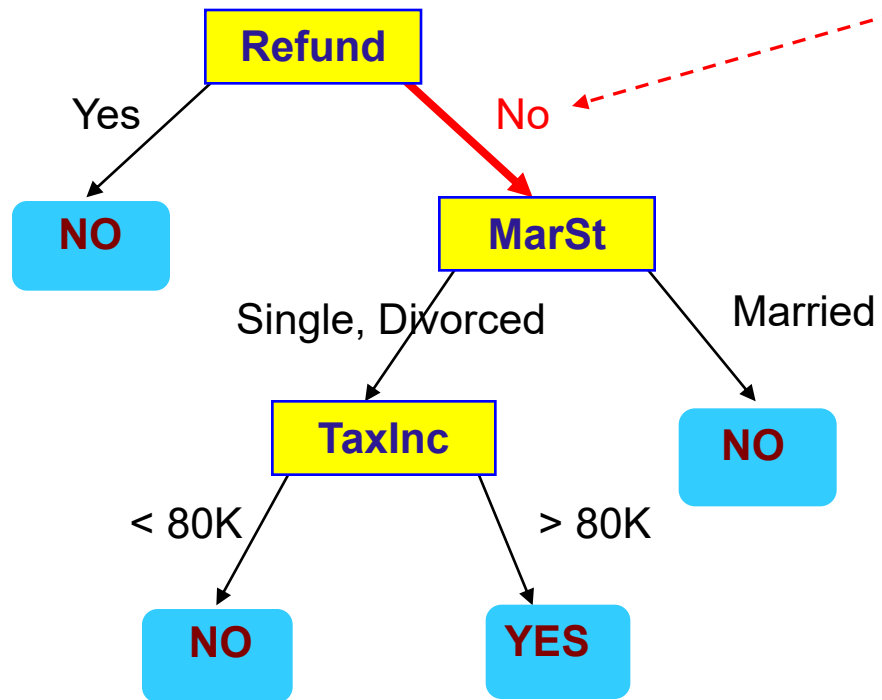
Apply Model to Test Data



Apply Model to Test Data

Test Data

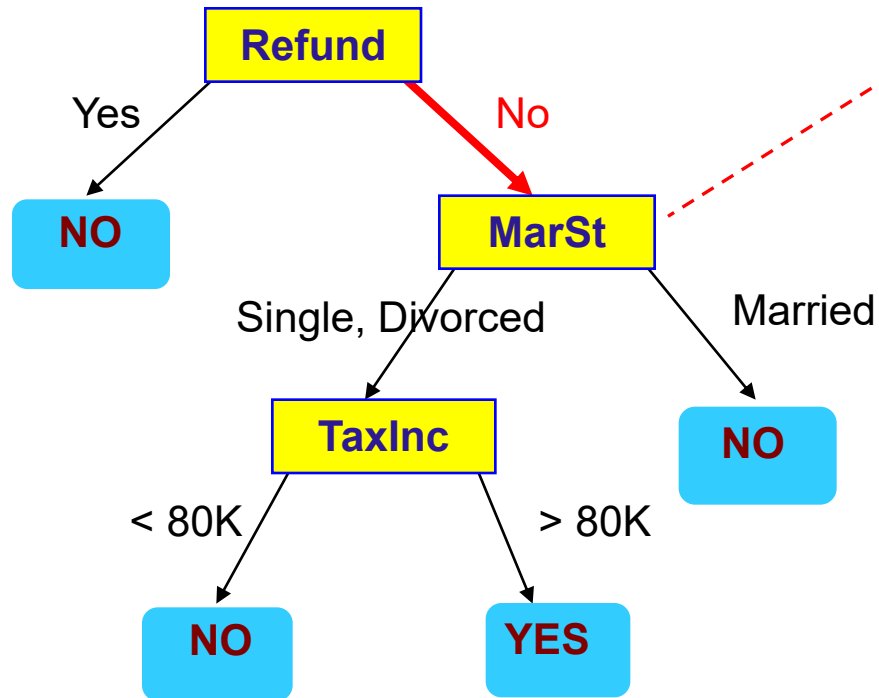
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



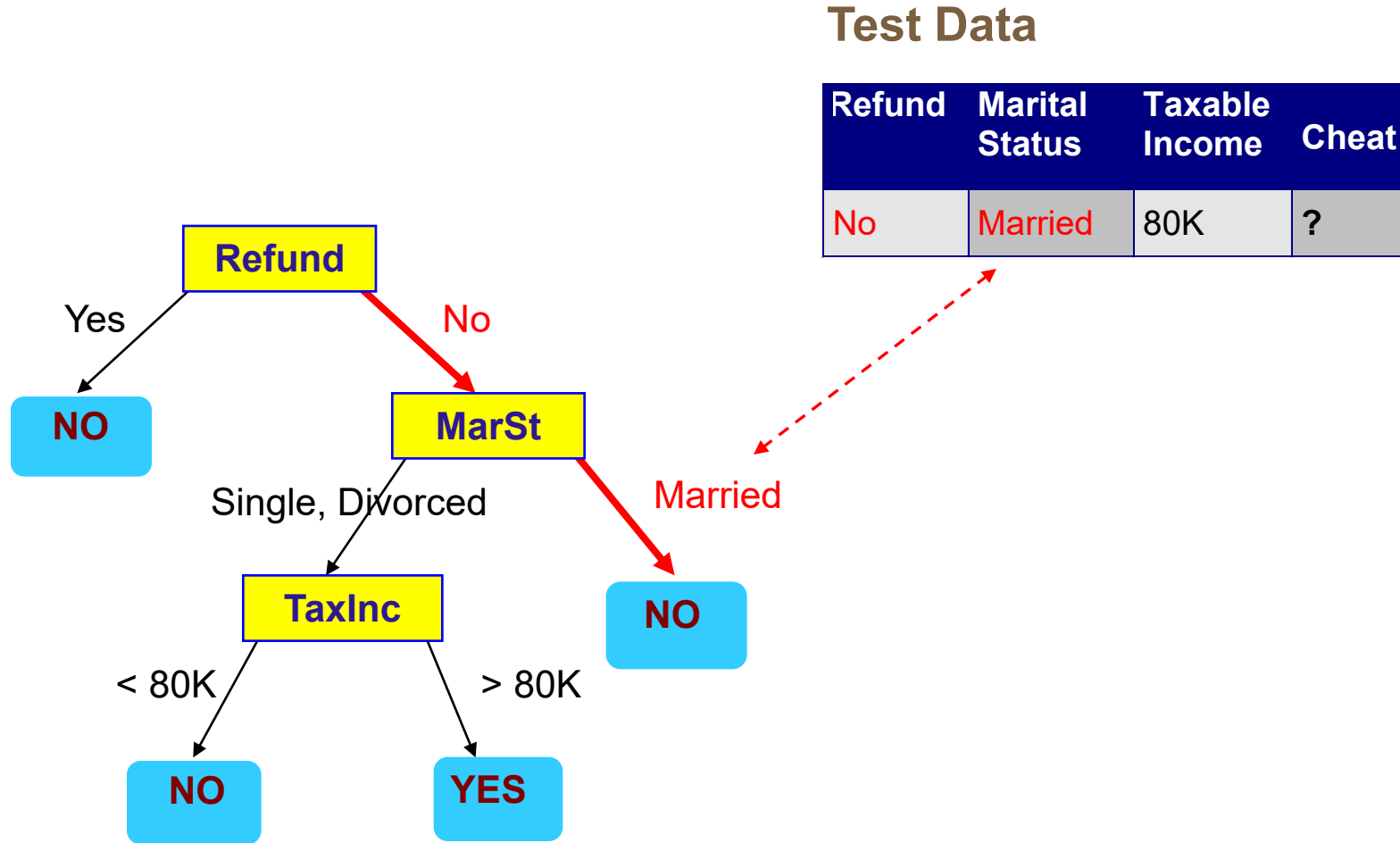
Apply Model to Test Data

Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



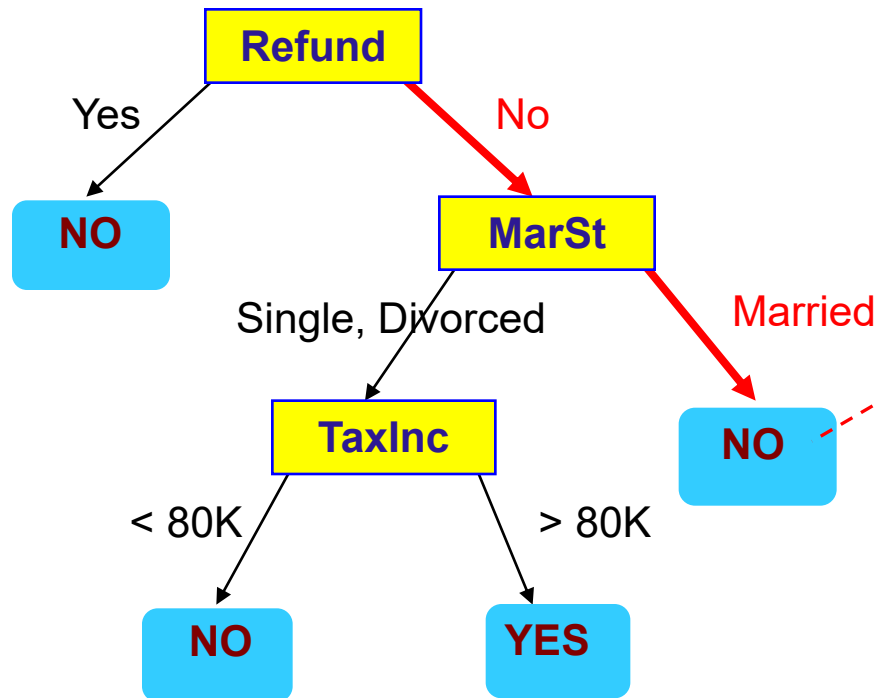
Apply Model to Test Data



Apply Model to Test Data

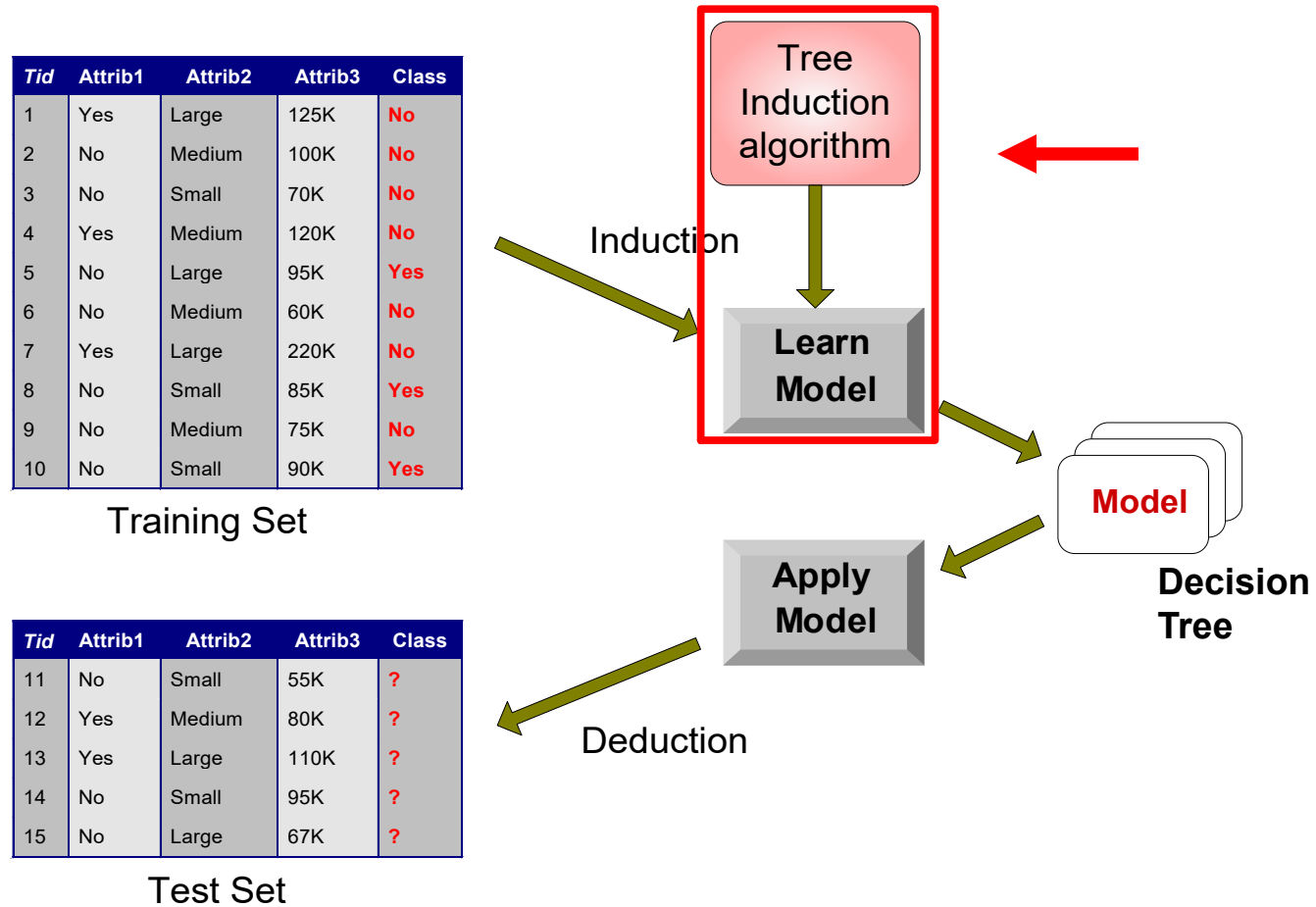
Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Assign Cheat to "No"

Decision Tree Classification Task



Decision Tree Induction

Many Algorithms:

- Hunt's Algorithm (one of the earliest)
- CART
- ID3, C4.5
- SLIQ, SPRINT



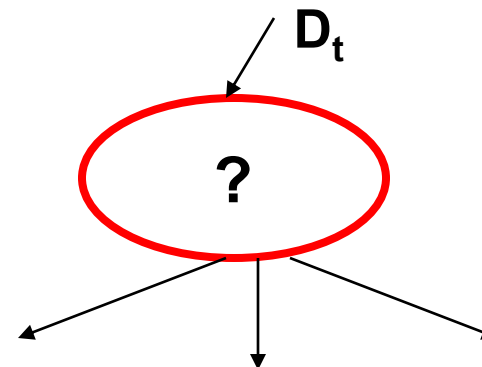
General Structure of Hunt's Algorithm

Let D_t be the set of training records that reach a node t

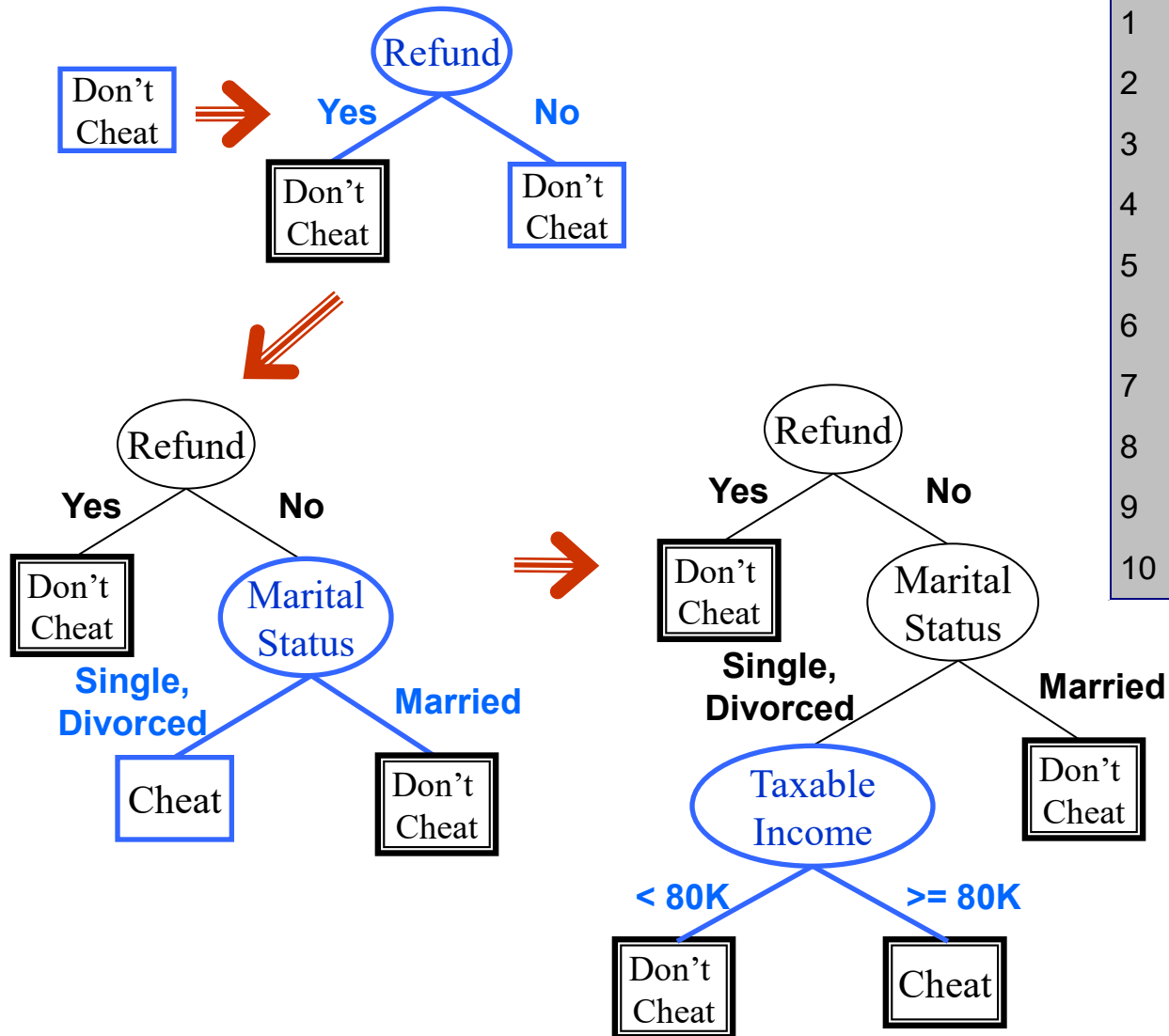
General Procedure:

- If D_t contains records that belong the same class y_t , then t is a leaf node labeled as y_t
- If D_t is an empty set, then t is a leaf node labeled by the default class, y_d
- If D_t contains records that belong to more than one class, use an attribute test to split the data into smaller subsets. Recursively apply the procedure to each subset.

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



Hunt's Algorithm



Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Tree Induction

Greedy strategy.

- Split the records based on an attribute test that optimizes certain criterion.

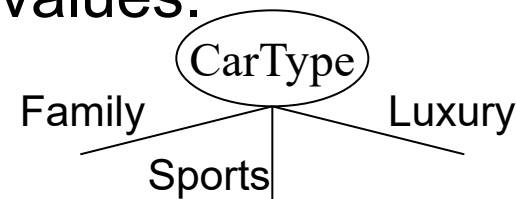
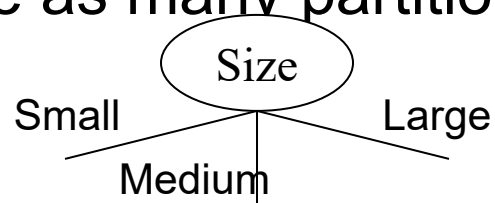
Issues

- Determine how to split the records
 - How to specify the attribute test condition?
 - **How to determine the best split?**
- Determine when to stop splitting



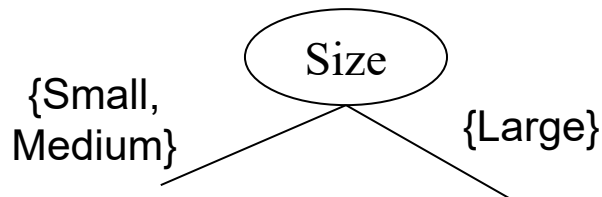
Splitting Based on Nominal and Ordinal Attributes

Multi-way split: Use as many partitions as distinct values.

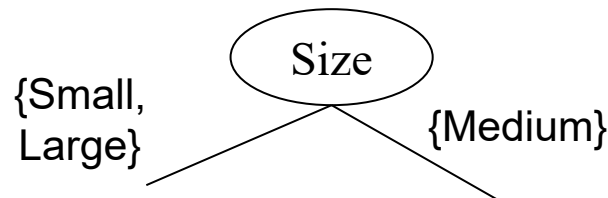
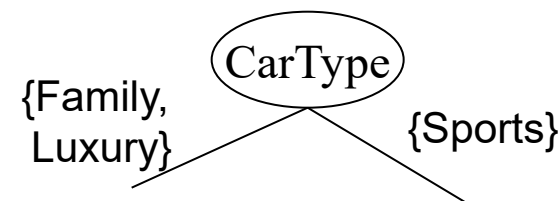


Binary split: Divides values into two subsets. Need to find optimal partitioning.

What about this split?



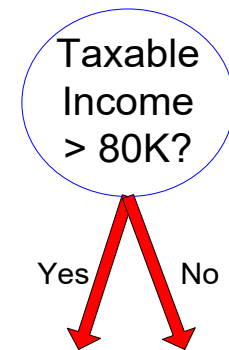
OR



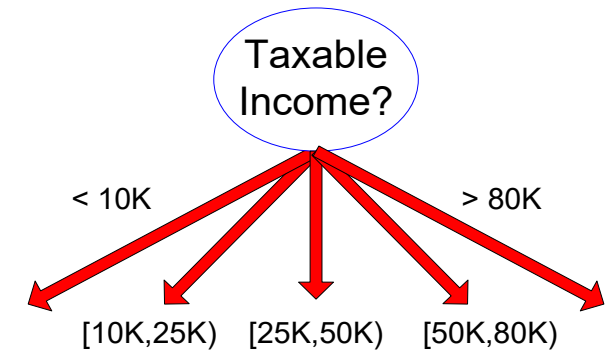
Splitting Based on Continuous Attributes

Different ways of handling

- **Discretization** to form an ordinal categorical attribute
 - Static – discretize once at the beginning
 - Dynamic – ranges can be found by equal interval bucketing, equal frequency bucketing (percentiles), or clustering.
- **Binary Decision**: $(A < v)$ or $(A \geq v)$
 - consider all possible splits and finds the best cut
 - can be more compute intensive



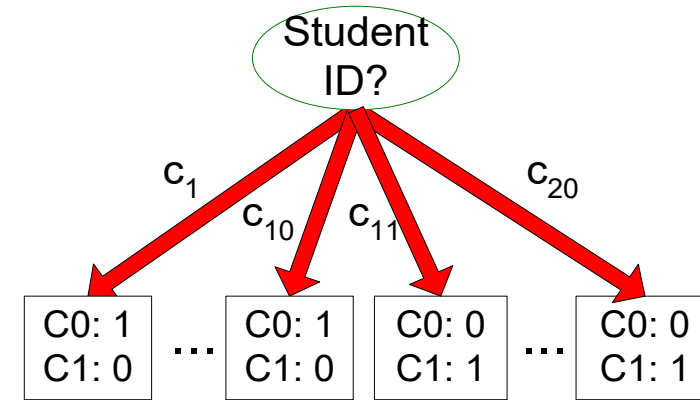
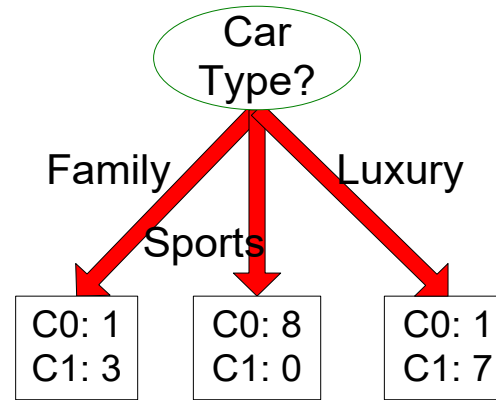
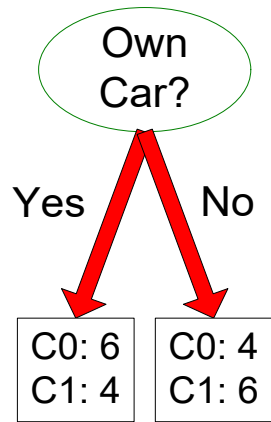
(i) Binary split



(ii) Multi-way split

How to determine the Best Split

Before Splitting: 10 records of class 0,
10 records of class 1



Which test condition is the best?

How to determine the Best Split

Greedy approach:

- Nodes with **homogeneous** class distribution are preferred

Need a measure of node impurity:

C0: 5
C1: 5

**Non-homogeneous,
High degree of impurity**

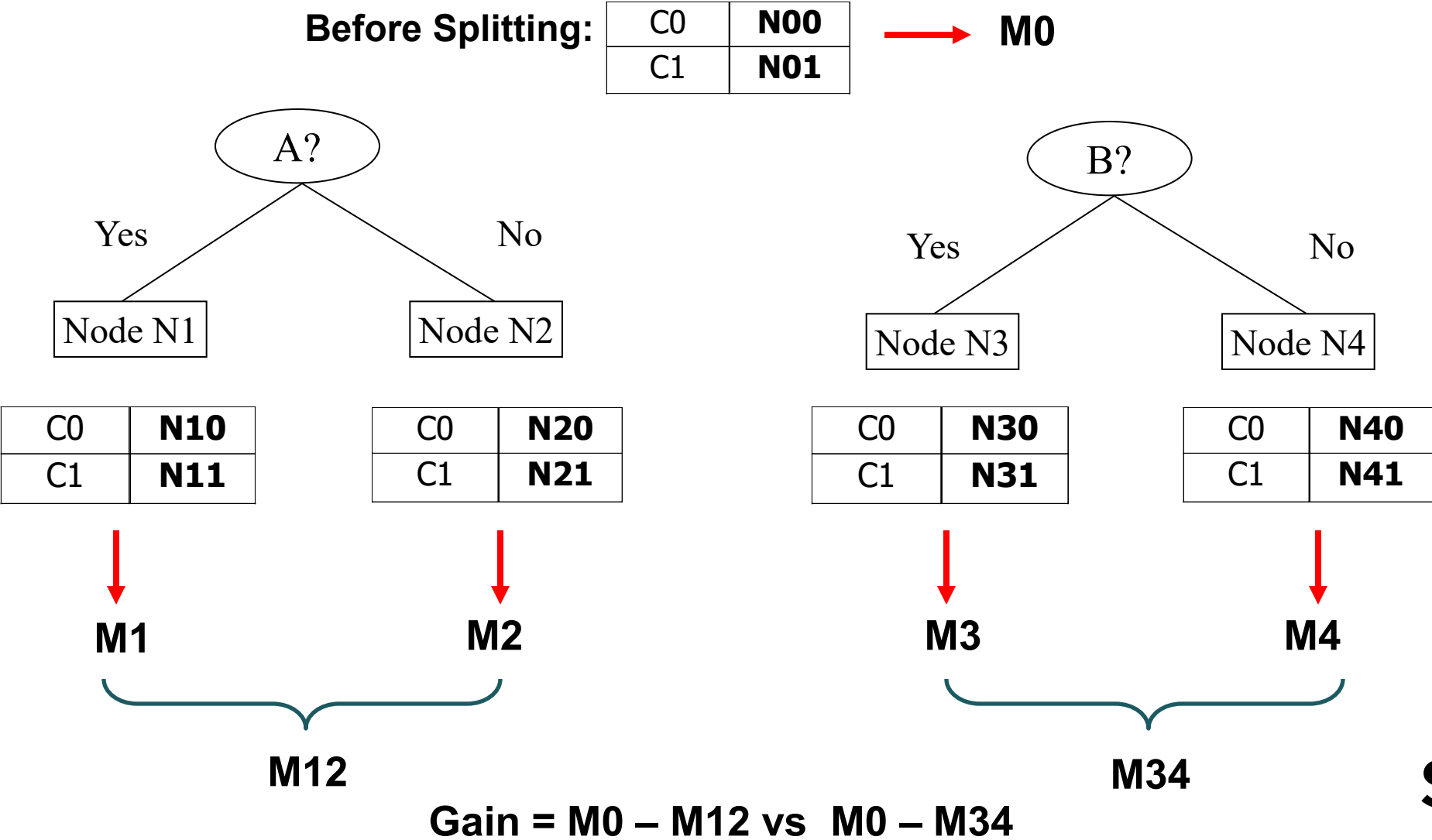
C0: 9
C1: 1

**Homogeneous,
Low degree of impurity**

Measures of Node Impurity

- ☐ Gini Index
- ☐ Entropy
- ☐ Misclassification error

How to Find the Best Split



Measure of Impurity: GINI

Gini Index for a given node t :

$$GINI(t) = 1 - \sum_j [p(j | t)]^2$$

(NOTE: $p(j | t)$ is the relative frequency of class j at node t).

- Maximum ($1 - 1/n_c$) when records are equally distributed among all classes, implying least interesting information
- Minimum (0.0) when all records belong to one class, implying most interesting information

C1	0
C2	6
Gini=0.000	

C1	1
C2	5
Gini=0.278	

C1	2
C2	4
Gini=0.444	

C1	3
C2	3
Gini=0.500	

Examples for computing GINI

$$GINI(t) = 1 - \sum_j [p(j | t)]^2$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$Gini = 1 - P(C1)^2 - P(C2)^2 = 1 - 0 - 1 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$Gini = 1 - (1/6)^2 - (5/6)^2 = 0.278$$

C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$Gini = 1 - (2/6)^2 - (4/6)^2 = 0.444$$

Splitting Based on GINI

Used in CART, SLIQ, SPRINT.

When a node p is split into k partitions (children), the quality of split is computed as,

$$GINI_{split} = \sum_{i=1}^k \frac{n_i}{n} GINI(i)$$

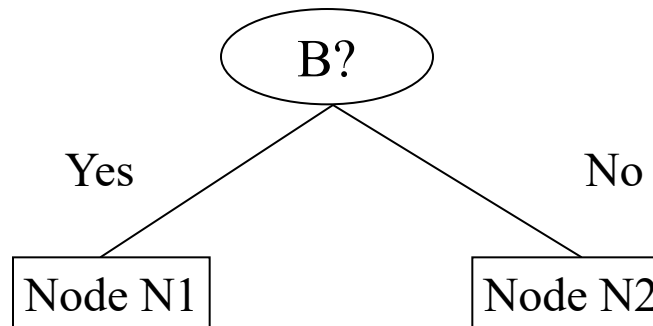
where,

n_i = number of records at child i ,

n = number of records at node p .

Binary Attributes: Computing GINI Index

- Splits into two partitions
- Effect of Weighing partitions:
 - Larger and Purer Partitions are sought for.



	Parent
C1	6
C2	6
Gini = 0.500	

$$\begin{aligned}\text{Gini(N1)} \\ &= 1 - (5/7)^2 - (2/7)^2 \\ &= 0.408\end{aligned}$$

$$\begin{aligned}\text{Gini(N2)} \\ &= 1 - (1/5)^2 - (4/5)^2 \\ &= 0.320\end{aligned}$$

	N1	N2
C1	5	1
C2	2	4
Gini=0.371		

$$\begin{aligned}\text{Gini(Children)} \\ &= 7/12 * 0.408 + \\ &\quad 5/12 * 0.320 \\ &= 0.371\end{aligned}$$

Continuous Attributes: Computing Gini Index

Use Binary Decisions based on one value

Several Choices for the splitting value

- Number of possible splitting values
= Number of distinct values

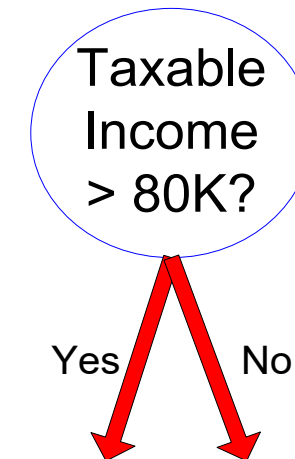
Each splitting value has a count matrix associated with it

- Class counts in each of the partitions, $A < v$ and $A \geq v$

Simple method to choose best v

- For each v , scan the database to gather count matrix and compute its Gini index
- Computationally Inefficient! Repetition of work.

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



Continuous Attributes: Computing Gini Index...

For efficient computation: for each attribute,

- Sort the attribute on values
- Linearly scan these values, each time updating the count matrix and computing gini index
- Choose the split position that has the least gini index

		Cheat	No		No		No		Yes		Yes		Yes		No		No		No		No			
		Taxable Income																						
Sorted Values	→	60		70		75		85		90		95		100		120		125		220				
		55		65		72		80		87		92		97		110		122		172		230		
		<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	
		Yes	0	3	0	3	0	3	0	3	1	2	2	1	3	0	3	0	3	0	3	0	3	0
		No	0	7	1	6	2	5	3	4	3	4	3	4	3	4	4	3	5	2	6	1	7	0
		Gini	0.420		0.400		0.375		0.343		0.417		0.400		<u>0.300</u>		0.343		0.375		0.400		0.420	

Alternative Splitting Criteria based on INFO

Entropy at a given node t:

$$Entropy(t) = -\sum_j p(j | t) \log p(j | t)$$

(NOTE: $p(j | t)$ is the relative frequency of class j at node t).

- Measures homogeneity of a node.
 - Maximum ($\log n_c$) when records are equally distributed among all classes implying least information
 - Minimum (0.0) when all records belong to one class, implying most information
- Entropy based computations are similar to the GINI index computations

Examples for computing Entropy

$$Entropy(t) = -\sum_j p(j | t) \log_2 p(j | t)$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$Entropy = -0 \log 0 - 1 \log 1 = -0 - 0 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$Entropy = - (1/6) \log_2 (1/6) - (5/6) \log_2 (1/6) = 0.65$$

C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$Entropy = - (2/6) \log_2 (2/6) - (4/6) \log_2 (4/6) = 0.92$$

Splitting Criteria based on Classification Error

Classification error at a node t :

$$Error(t) = 1 - \max_i P(i | t)$$

Measures misclassification error made by a node.

- Maximum ($1 - 1/n_c$) when records are equally distributed among all classes, implying least interesting information
- Minimum (0.0) when all records belong to one class, implying most interesting information

Examples for Computing Error

$$Error(t) = 1 - \max_i P(i | t)$$

C1	0
C2	6

$$P(C1) = 0/6 = 0 \quad P(C2) = 6/6 = 1$$

$$Error = 1 - \max(0, 1) = 1 - 1 = 0$$

C1	1
C2	5

$$P(C1) = 1/6 \quad P(C2) = 5/6$$

$$Error = 1 - \max(1/6, 5/6) = 1 - 5/6 = 1/6$$

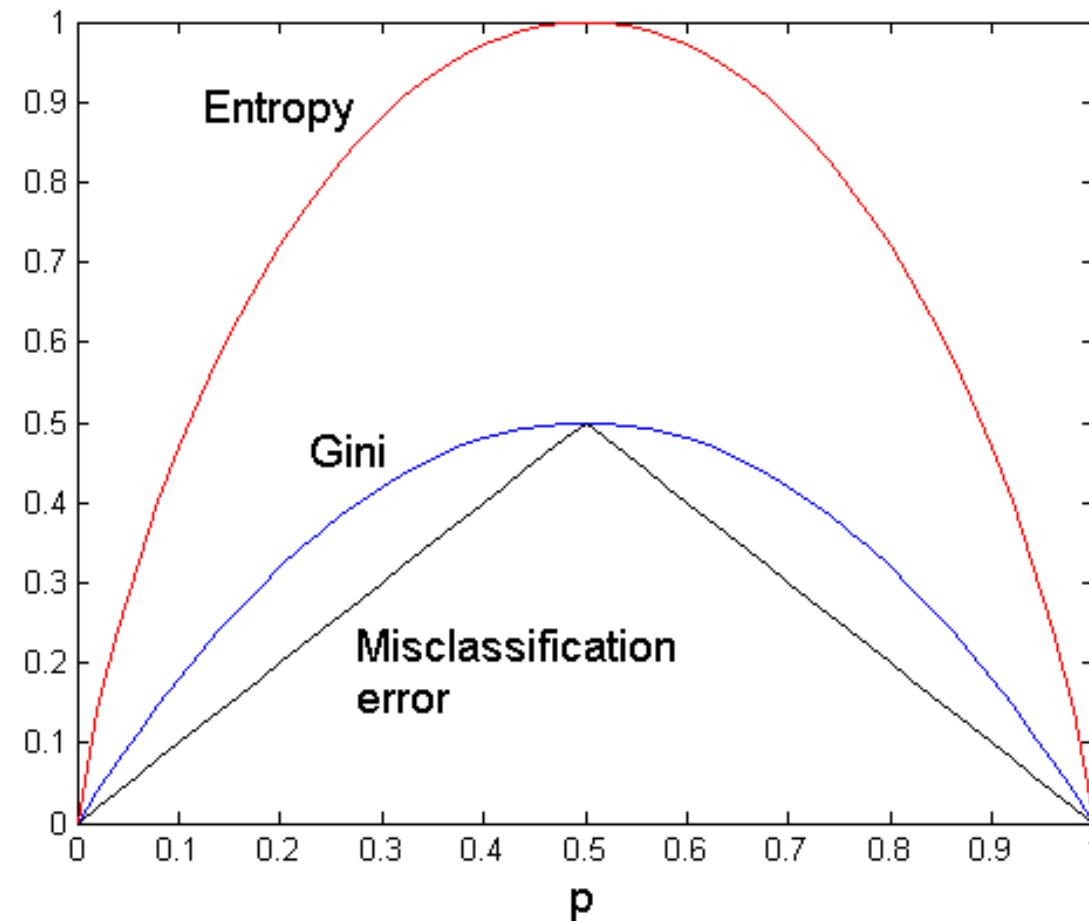
C1	2
C2	4

$$P(C1) = 2/6 \quad P(C2) = 4/6$$

$$Error = 1 - \max(2/6, 4/6) = 1 - 4/6 = 1/3$$

Comparison among Splitting Criteria

For a 2-class problem:



Stopping Criteria for Tree Induction



Stop expanding a node when all the records belong to the same class



Stop expanding a node when all the records have similar attribute values



Early termination

using hyperparameters such as
max_depth,
min_samples_leaf, min_samples_split,
etc.

Decision Tree Based Classification



Advantages:

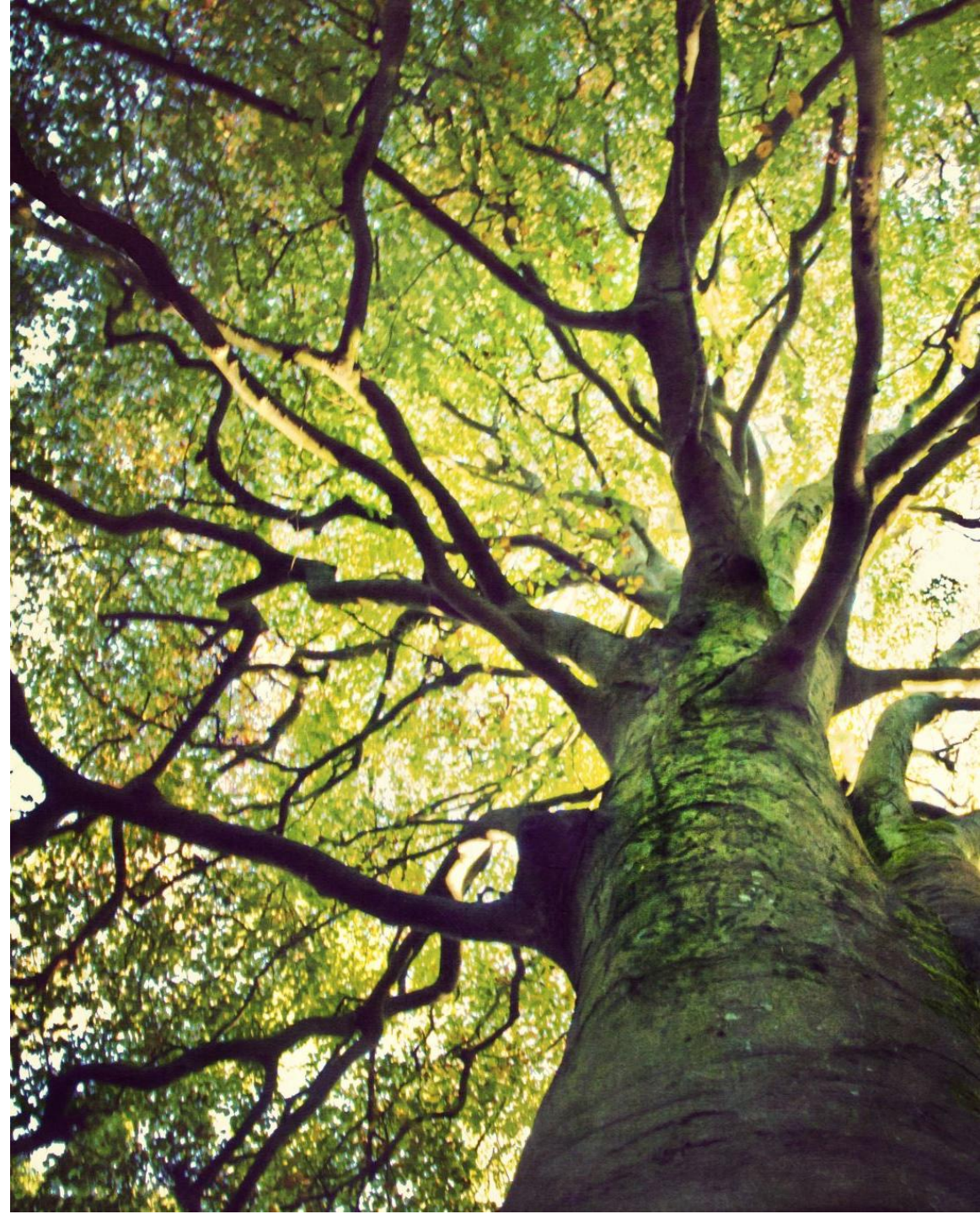
- Inexpensive to construct
- Extremely fast at classifying unknown records
- Easy to interpret for small-sized trees
- Accuracy is comparable to other classification techniques for many simple data sets

Practical Issues of Classification

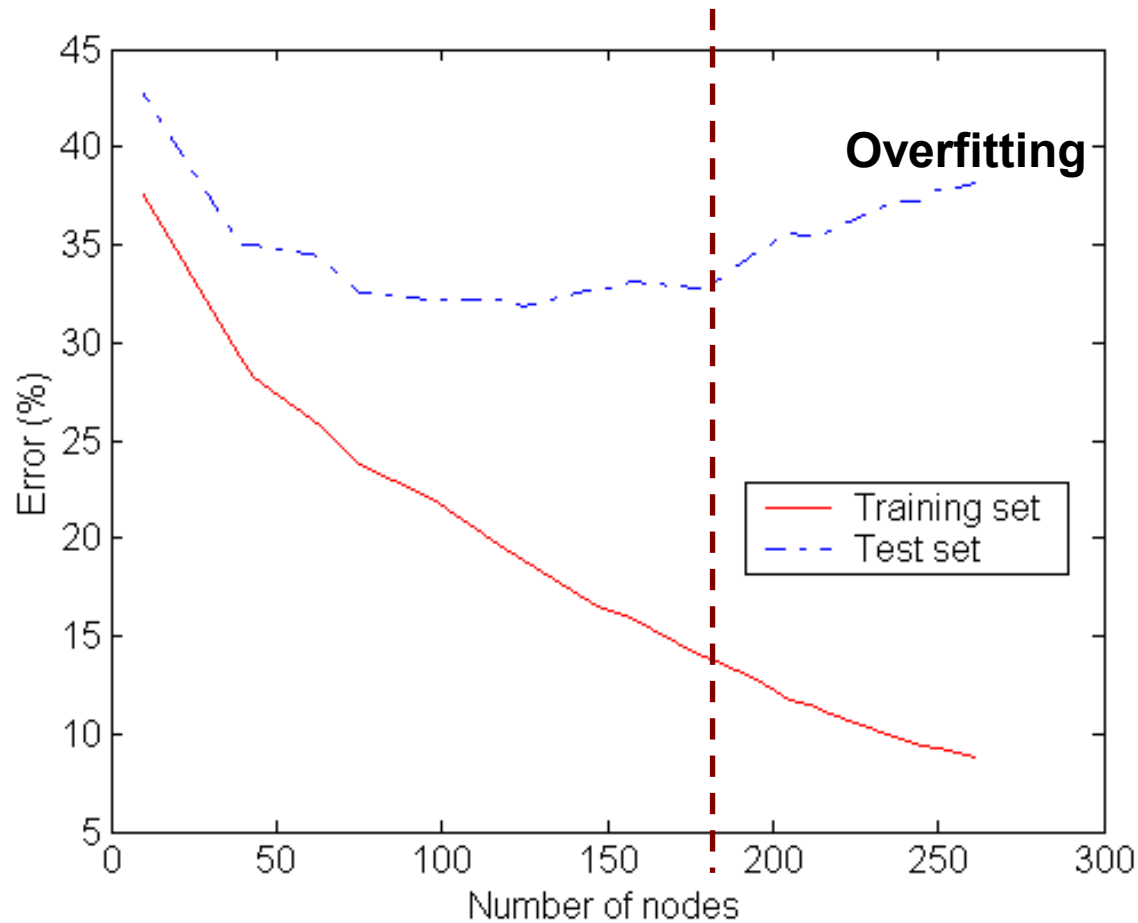
Underfitting and Overfitting

Missing Values

Costs of Classification



Underfitting and Overfitting

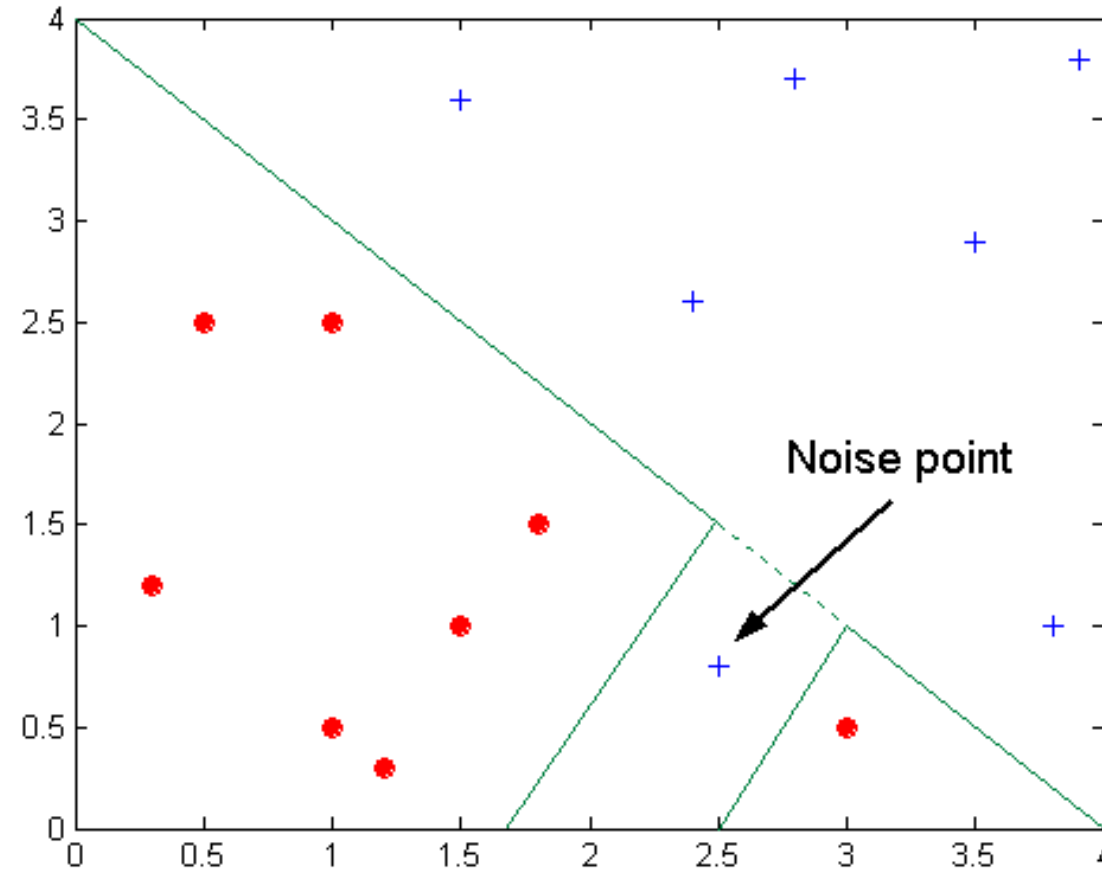


Overfitting: when model is more complex than necessary.

Training error no longer provides a good estimate of how well the tree will perform on unseen records

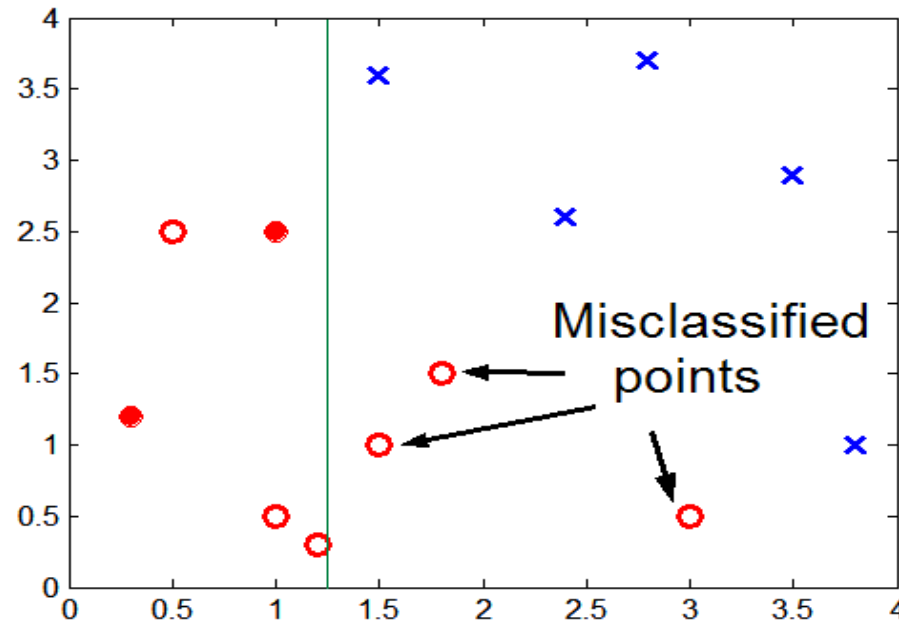
Underfitting: when model is too simple, both training and test errors are large

Overfitting due to Noise



Decision boundary is distorted by noise point

Overfitting due to Insufficient Examples



-Lack of data points in the lower half of the diagram makes it difficult to predict correctly the class labels of that region

-Insufficient number of training records in the region causes the decision tree to predict the test examples using other training records that are irrelevant to the classification task

How to Address Overfitting

Pre-Pruning (Early Stopping Rule)

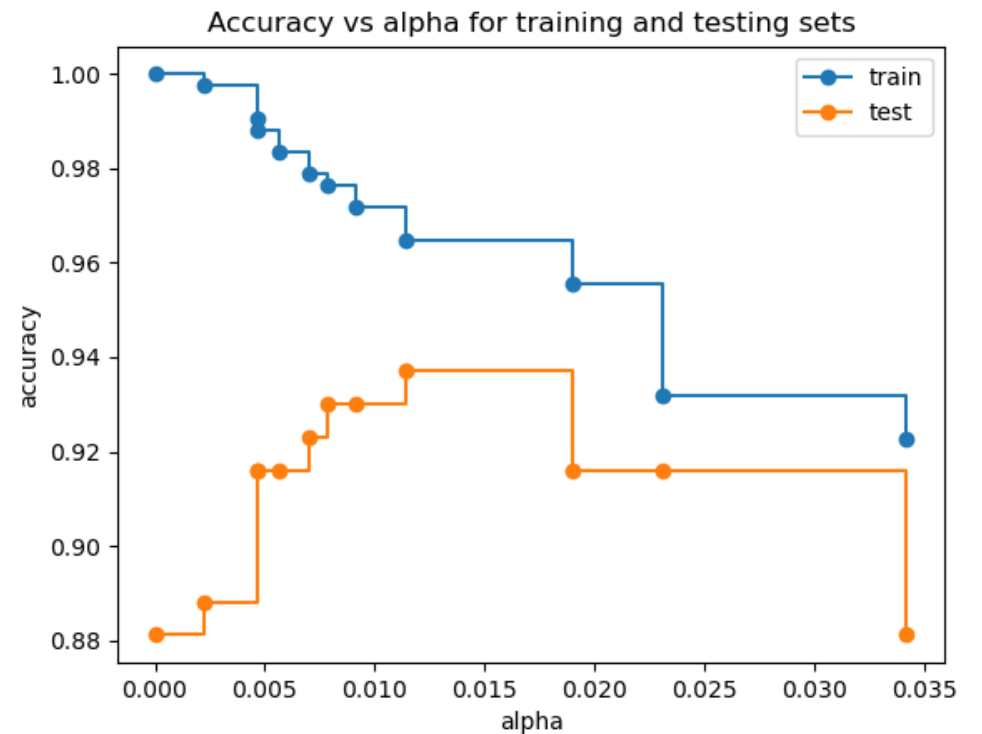
- Stop the algorithm before it becomes a fully-grown tree
- Typical stopping conditions for a node:
 - Stop if all instances belong to the same class
 - Stop if all the attribute values are the same
- More restrictive conditions:
 - Stop if number of instances is less than some user-specified threshold
 - Stop if class distribution of instances are independent of the available features (e.g., using χ^2 test)
 - Stop if expanding the current node does not improve impurity measures (e.g., Gini or information gain).



How to Address Overfitting...

Post-pruning

- Grow decision tree to its entirety
- Trim the nodes of the decision tree in a bottom-up fashion
- If generalization error improves after trimming, replace sub-tree by a leaf node.
- Class label of leaf node is determined from majority class of instances in the sub-tree
- Can use MDL or Cost complexity pruning (ccp-alpha) for post-pruning



**Thanks for your
attention**